CS/W-24/CS3T003/02

Education to Eternity

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

॥ज्ञानम् सर्वार्थ साधनम्॥

B. Tech ~~Summer~~ / Winter - 2024

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: Computer Science & Engineering **Sem:- III**

Subject: Computer Architecture & Organization **Subject Code: CS3T003**

Marks: 100 **Date of Exam: 19/11/2024** **Time: 3 Hrs.**

Instructions to the students:

1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicated.**
3. **Assume suitable data whenever necessary.**
4. **Illustrate your answer whenever necessary with the help of neat sketches.**
5. **Use of Programmable Calculator is prohibited.**
6. **Don't use red pen for writing the answers.**

COs: Students should be able to:

1. Summarize the organization and operation of digital computers.
2. Study and apply knowledge of processor instruction sets and its execution.
3. Demonstrate computer arithmetic operations on integer and floating-point numbers.
4. Describe the organization of memory system.
5. Explain concepts of I/O organization and pipelining of a processor.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one/two word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Number System.	L1/CO1	2
Q.2	State the difference between RAM and ROM.	L2/CO1	2
Q.3	Write short note on Processor.	L3/CO5	2
Q.4	List out types of Computer Organization.	L1/CO1	2
Q.5	Rewrite about Cache Memory.	L2/CO4	2
Q.6	Classify the types of Interleaved Memory.	L2/CO4	2
Q.7	Restate the types of Buses.	L2/CO5	2
Q.8	Write the Mapping techniques.	L3/CO4	2
Q.9	Give example of hardware and software.	L2/CO1	2
Q.10	Define Interleaved Memory.	L1/CO4	2
Q.11	State the Advantages of Direct-mapping.	L1/CO5	2
Q.12	Sketch microinstruction code format.	L1/CO3	2

SECTION B

1. **All Questions are compulsory.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Distinguish between virtual memory and cache memory.	L4/CO2	4
Q.14	Illustrate different addressing modes in Computer Architecture	L3/CO1	4
Q.15	Discuss hardwired and microprogrammed control unit.	L2/CO3	4
Q.16	Examine X=(A+B)*(C-D)/E using 3, 2 and 0 Address Instruction format.	L4/CO2	4
Q.17	Write short note on Booth's Algorithm.	L3/CO2	4
Q.18	Sketch single and double precision format of IEEE floating point representation.	L3/CO2	4

SECTION C

1. **Solve any FOUR Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Illustrate memory management requirements.	L4/CO4	7
Q.20	Explain non restoring algorithm with example.	L5/CO2	7
Q.21	Analyse RISC & CISC with characteristics.	L4/CO4	7
Q.22	Examine multiplication of two signed number $(10111)_2$ & $(10011)_2$ by suitable algorithm	L4/CO2	7
Q.23	Describe parallel processor system in computer architecture.	L5/CO5	7
Q.24	Discuss the concept of pipelining. How it improves the execution of program over sequential execution?	L5/CO5	7

SECTION D

1. **Solve any TWO Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Write short note on following: a) Programmed IO. b) Vector processing. c) IO modules. d) Control unit	L5/CO5,2	12
Q.26	Evaluate division of 17 and 3 by using restoring algorithm.	L6/CO2	12
Q.27	Describe Direct-mapping, Associative Mapping & Set-Associative Mapping.	L6/CO5	12